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CS-499
Enhancement Three Narrative

This artifact is the same as the previous two artifacts, which is the mobile inventory management application developed for the Android OS. It was created in a previous course at SNHU, CS-360 Mobile Architecture and Programming. The application allows users to create an account, which they must do before logging in. After the user successfully creates an account and logs in, the app will navigate to the inventory screen, where they can view the items in the inventory, add an item, modify item attributes, and delete items. The application uses Room to store data for user accounts and for inventory items.

I selected this artifact for my ePortfolio because the current implementation of the database lacks several aspects that would make the implementation acceptable in a professional application, making it a prime candidate for enhancement. The artifact was improved in a few ways. First, input validation for the database operations were improved. Validation was added in the repository and application layers to ensure that duplicate items are not added to the database. Secondly, the database schema for the user entity was changed to eliminate the storage of plain-text passwords. The password column was replaced with a password hash and salt. This updated the database to abide by security best practices of salting and hashing user passwords, rather than storing secrets in plain text. Finally, a Room migration was added for the inventory items. The Room database previously used fallbackToDestructiveMigration(), which erases all stored data when the schema changes. A migration was used to update the inventory item entity to include category and low-stock threshold attributes for the inventory items. This showcased my skills in having a security mindset, incorporating security by design, and prioritizing data quality and integrity.

I did meet the course outcomes I planned to meet in the module one materials, which were course outcomes four and five. Course outcome four was met by using well-founded techniques and tools to provide the database migration, as well as well-founded security packages that improved the security of the application's sensitive data. Course outcome five was met in that the enhancement demonstrated a security-focused mindset by implementing input validation across multiple layers of the application, as well as eliminating the security flaw of storing plain-text passwords in the database. I do not have any updates to the outcome coverage plan at this time.

The process of enhancing this artifact provided a learning experience in a couple different ways. I learned the importance of using input validation across multiple layers of the application. I had the unique constraint already applied to the name attribute of the inventory item, but initially I did not have any validation to catch errors if an attempt was made to create a duplicate item. The unique constraint worked at preventing the item's insertion, but the program would crash. Properly handling the error at the application level prevented crashes and improved the user experience. I also learned how to use some aspects of the security packages used to generate a salt for the password as well as hash it. Finally, I learned that setting up a Room migration is not a catch-all for all future schema changes. Rather it is a targeted path for changing the schema between specific versions, where the specific changes are denoted in the migration plan.